The concern over the possible negative effects of AI on income inequality and labor market polarization has grown with the pace of AI adoption. A report by McKinsey Global Institute found that generative AI could automate tasks that are similar to 60–70 percent of employees' existing work activities. The report found that the exposure to AI automation is the greatest in developed economies. The World Bank found that automation transforms occupations by altering task composition rather than displacing workers (World Bank). In a task-based production structure, AI is used to replace certain tasks, thereby altering labor demand. The economic issue is not whether AI transforms labor demand; it is whether the benefits of increased productivity due to AI adoption are shared among all labor types or are skewed towards high-skill labor.

Moreover, the macroeconomic impact of AI on the economy is significant. According to PwC, AI could contribute up to $15.7 trillion to the world's GDP by 2030 through productivity and consumer demand effects. However, as we have seen from the past, this does not necessarily mean that wages will keep rising. According to research from the Federal Reserve Bank of St. Louis, labor productivity in the United States has risen significantly faster than the median real wages of production and nonsupervisory workers since the 1970s, which has led to rising income inequality. This is because the labor share of income in the United States has been falling. The gap could widen further if AI continues to drive productivity and complements skilled labor and capital.

The distribution of the impact of AI is not even over the skill axis. According to the World Economic Forum, jobs that involve routine cognitive tasks, such as data entry, accounting, and clerical work, are at a high probability of being impacted by the increase in technology. These are the roles, particularly in the U.S., which have constantly offered middle-income jobs without requiring the workers to possess high educational qualifications. Even with the increase in the ability of AI to perform language generation, prediction, and pattern recognition, this transition from capital to labor will push the demand inward and thereby affect workers in these roles. Task switching may not necessarily lead to the replacement of workers in these roles, but the history of those who have been impacted by past automation does not paint a rosy picture.

Significantly, by contrast, high-skill labor in technical, managerial, and analytical fields appears to benefit from AI in a complementary fashion. If AI helps with complex problem solving and strategic decisions, it complements the skills of highly educated workers by raising their marginal productivity. If AI capital augments high-skill labor, the technology improves the demand for the labor, which attracts corresponding wages and enhances the return from investment in education.

Low-skill industries such as caregiving, food service, and hospitality are unlikely to benefit from full automation due to the need for physical dexterity and people management skills. However, these industries also see low productivity growth. If monopsony power and low bargaining is a

feature of these service industries in the US market, such as the ones described above, then the labor market is unlikely to see a rise in wages despite the rise of AI. This is consistent with the job polarization results from the National Bureau of Economic Research, which suggests that computerization is leading to more jobs being created at the low- and high-wage extremes and fewer middle-wage jobs (Autor and Dorn). Cognitive labor might see more pressure from the rise of AI as middle-skills jobs might see more disruption from AI compared to the past.

One of the many options is unconditional income security in the form of unconditional cash transfers, otherwise known as Universal Basic Income (UBI). In an article by economists Damon Jones and Ioana Marinescu about an investigation of Alaska's Permanent Fund Dividend, they discovered that when people receive unconditional annual cash transfers, employment levels don't take a hit; on the contrary, people take up part-time jobs. The Finnish trial of universal basic income, as discovered by Kela, saw recipients of universal cash transfers experience better psychological well-being as well as slight labor-related effects.

In an AI disruption-driven economy, income security programs would help alleviate risk aversion as well as liquidity problems, permitting workers who have lost their jobs to undergo new training, move to other locations, or become entrepreneurs. The programs would also help moderate the demand shock that might occur in areas that are particularly adversely affected by the loss of employment.

Funds to support such programs, however, pose important questions of both fiscal and efficiency implications. For instance, the transfer would have to be substantial to compensate for displaced workers and would thus necessitate higher federal expenditure and perhaps the imposition of further taxes, or the diversion of already allocated public expenditure. While this strategy of providing UBI might address income inequality, it might not directly alleviate skill deficiencies or the corresponding fall of demand for cognitive labour. Thus, it might not reverse the underlying phenomenon of wage polarization, and other options, such as wage insurance, might present better options and fiscal precision, despite their associated difficulties and possible disincentives to work. In other words, income security might alleviate transitional pains and politics, but long-term success rests with other, more important, reforms, which will shape how gains from AI are equitably distributed.

Ultimately, AI could have a positive impact on productivity levels and labor demand patterns in uneven ways among American laborers. High-skilled workers could experience higher productivity levels and wages if they complement AI; meanwhile, the labor market could witness downward mobility among middle-skill laborers due to the displacement of routine cognitive tasks from the labor force. Low-skill laborers could be fortunate to keep their jobs with moderate wages. rate this paper

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